

Gelucire 48/16® for Type-IV Lipid Based Formulations and Their *In Vitro In Vivo* Performance Evaluation

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PURPOSE

- Ticagrelor (TGR) is an oral antiplatelet agent, indicated to reduce the rate of thrombotic cardiovascular events.
- A BCS Class-IV drug attributing to its poor solubility and permeability with log P value 2.
- Gelucire 48/16® was evaluated to improve solubility and bioavailability of TGR.
- Gelucire 48/16® is a PEG 32 mono and diesters of stearic and palmitic acid, with HLB of 16 and CMC of 153±31 mg/L which helps to increase solubility of hydrophobic and lipophobic drugs
- Being a solid at ambient temperature with a melting point of 48°C, makes it amenable for capsule filling, melt processes and tableting

Gelucire® 48/16: Solubilizer for BCS Class 4 Drugs	
Pharmacopoeial description:	
- Polyoxyl stearate (Type I) USP-NF - Polyethylene glycol monostearate (JPE) - Macrogol stearate (Type I) EP (pending)	
Physico-chemical information	
Trade name	Gelucire® 48/16
Drop point	48.4 ± 0.4
Melting point onset (DSC)	45.4 ± 0.2°C
pH (10% in purified water)	5 ± 1
HLB calculated/practical	16/12
Hydrophilic component	PEG 1500
Critical Micellar Concentration (mg/L) (CMC) at 25°C	153 ± 31
Particle size (37°C, 1g/200mL water min)	7 ± 1
Form	Solid (pellets)
Compatible with hard shell gelatin (ConiSnap), HPMC capsules HPMC (V Caps) and soft gelatin capsule.	

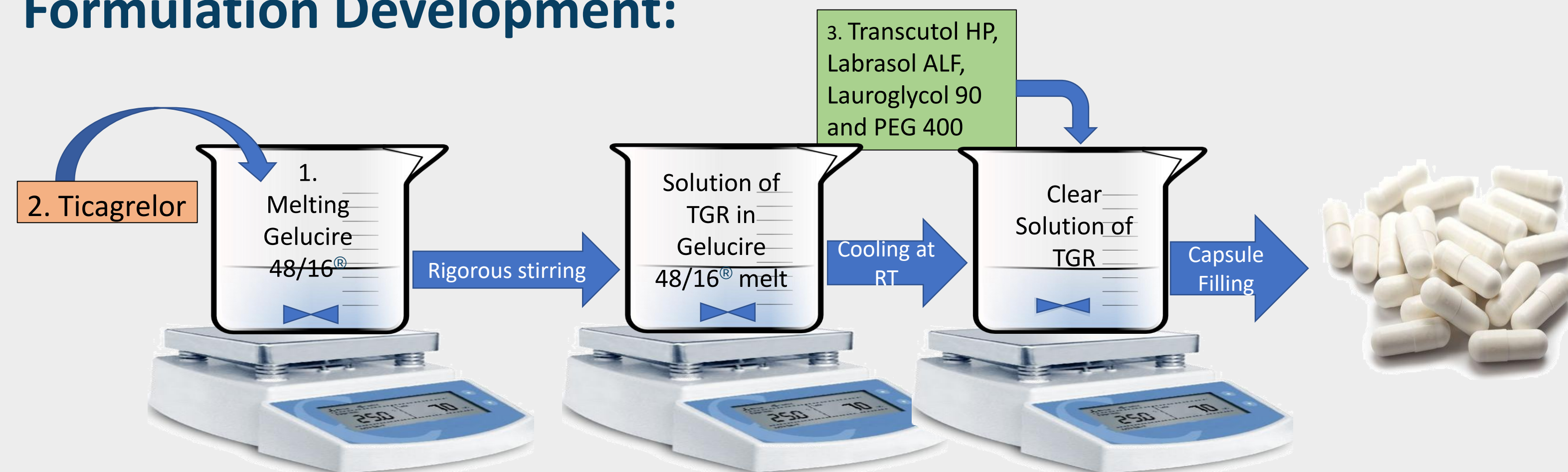
OBJECTIVES

- To prepare Type 4 lipidic formulation of TGR with Gelucire 48/16® by hot melt technique to improve solubility of TGR.
- To convert the formulation of TGR with Gelucire 48/16® in a capsule dosage form; from a suprageneric formulation (505 b ii) perspective.
- To evaluate the developed formulations for their performance by *in vitro* lipolysis study and *in vivo* Pharmacokinetic study on rats and comparison with market reference.

METHODS

Solubility Studies: Solubility of TGR was evaluated in water along with its dispersions with Gelucire 48/16® in various ratios. The samples were stirred for 72h at 37±0.2°C and evaluated for drug content in supernatant after centrifugation. TGR showed the affinity towards Gelucire 48/16® to enhance its solubility.

Formulation Development:



Step-1: Gelucire 48/16® was melted at 70°C

Step-2: TGR sieved through #40 and added in the molten mass of Gelucire 48/16® under constant stirring-TGR solubilizes

Step-3: The solution of TGR in Gelucire 48/16® cooled till 40°C followed by addition of Transcutol HP (Solvent), Labrasol ALF (Cosurfactant), Lauroglycol 90 (Cosolvent) and PEG 400 (Cosurfactant). Three formulations were fabricated with different compositions.

Step-4: The solution of Step 3 was filled into hard gelatin capsules and kept for cooling at RT.

The capsule formulations were evaluated for *in vitro* dissolution tests. *In vitro* lipolysis test and *in vivo* Pharmacokinetic study on rat model.

RESULTS

In vitro Dissolution:

- Dissolution of developed formulations and market reference was performed in 900 mL of dissolution media i.e. HCl Buffer pH 1.2, Acetate Buffer pH 4.5, Phosphate Buffer pH 6.8, and 0.2% Polysorbate 80, using USP-II apparatus, 75 rpm at 37°C.
- The dissolution of the developed lipidic formulation was found to be increased remarkably compare to market reference (Fig. 1).

In vitro Lipolysis Study:

- Apparatus:** pH stat apparatus (Metrohm AG, Switzerland) (Fig. 2),
- Medium:** Lipolysis medium pH 6.5 (fasted state): Tris Maleate 2mM, CaCl₂ 2H₂O 1.4 mM, NaCl 150 mM, NaTDC 3mM, Phosphatidyl choline 0.75 mM, MiliQ water QS.
- Volume:** 36 mL
- Gelucire 48/16® showed very good solubilization capacity. All three formulations demonstrated good capacity to maintain TGR in solution upon digestion of lipids. Formulation 1 and 2 were taken for *in vivo* pharmacokinetic study (Table 1, Fig. 3).

In vivo Pharmacokinetic Study:

- Animal Model:** Wistar male rats with weight of 200 to 250 g
- Dose:** 10 mg of TGR per kg body weight
- Route of Administration:** Oral gavage 1 to 1.5 mL
- No. of Groups:** 2 groups for GIPL formulations; 1 for Market reference
- Sampling:** Blood samples were collected at predefined time intervals up to 24 h through retro-orbital plexus under anaesthesia.
- Analysis:** The plasma was separated and TGR was analyzed using a validated HPLC method.

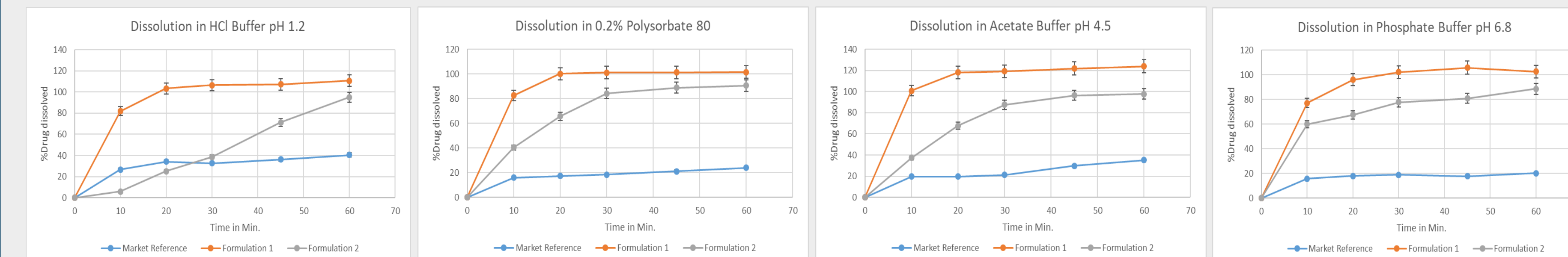


Fig. 1: *In vitro* dissolution of lipidic formulations and market reference in various dissolution media.



Fig. 2: pH Stat Apparatus.

Table 1: *In vitro* Lipolysis study of GIPL formulations:

Parameters	F-1	F-2	F-3
Dose (mg)	90	90	90
Fill weight (mg)	900	750	600
% TGR dissolved in First 5 min	97.2	98	98.6
AUC (mg*min)	4760	4233	3909

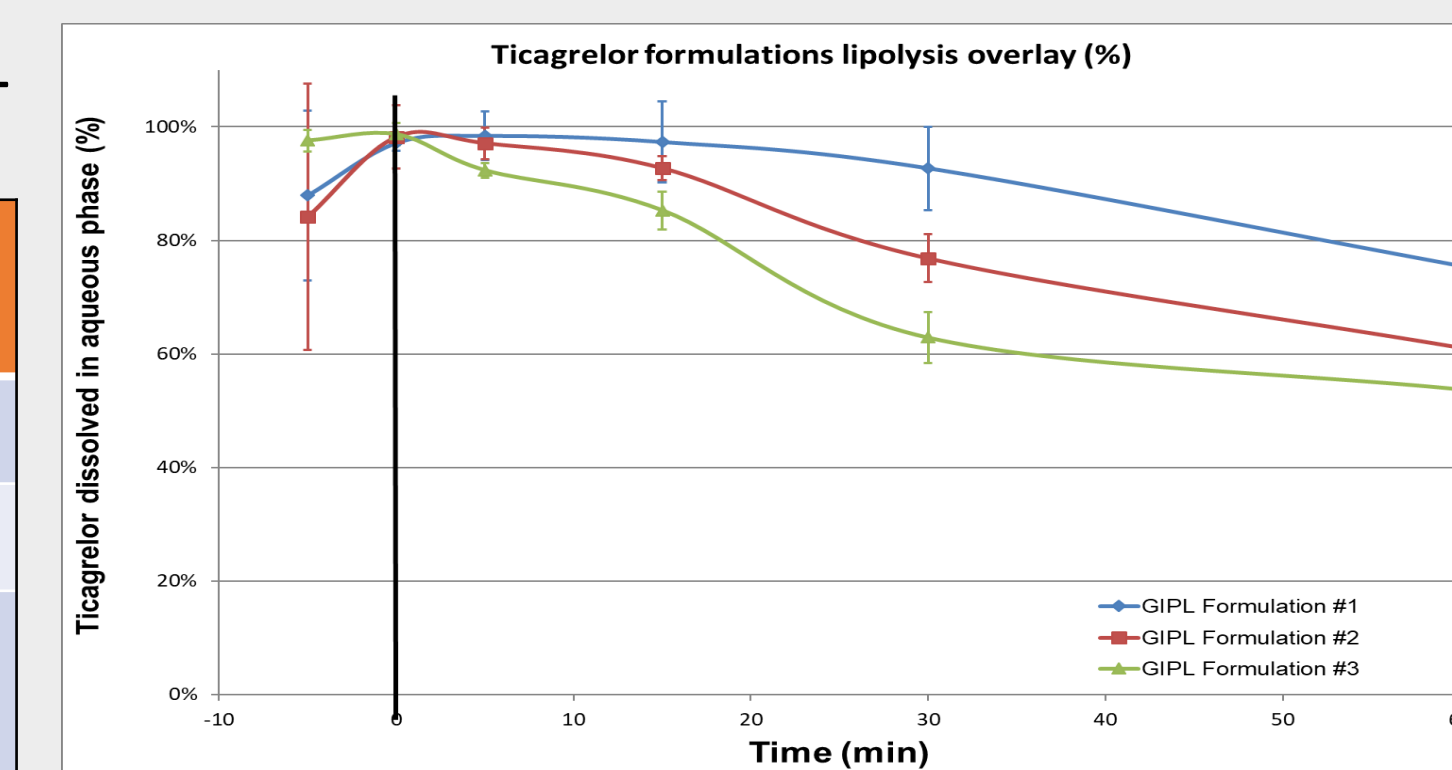


Fig. 3: *In vitro* Lipolysis study of GIPL formulations:

Table 2: *In vivo* Pharmacokinetic Study on Rats:

Parameters	Formulation 1	Formulation 2	Market reference
T _{max} (h)	1.50	2.15	3.54
C _{max} (ng/mL)	343.74	434.69	214.52
AUC _{0-t} (ng/mL*h)	4340.92	4530.75	2678.64
AUC _{0-inf} (ng/mL*h)	5309.66	4882.35	2972.53

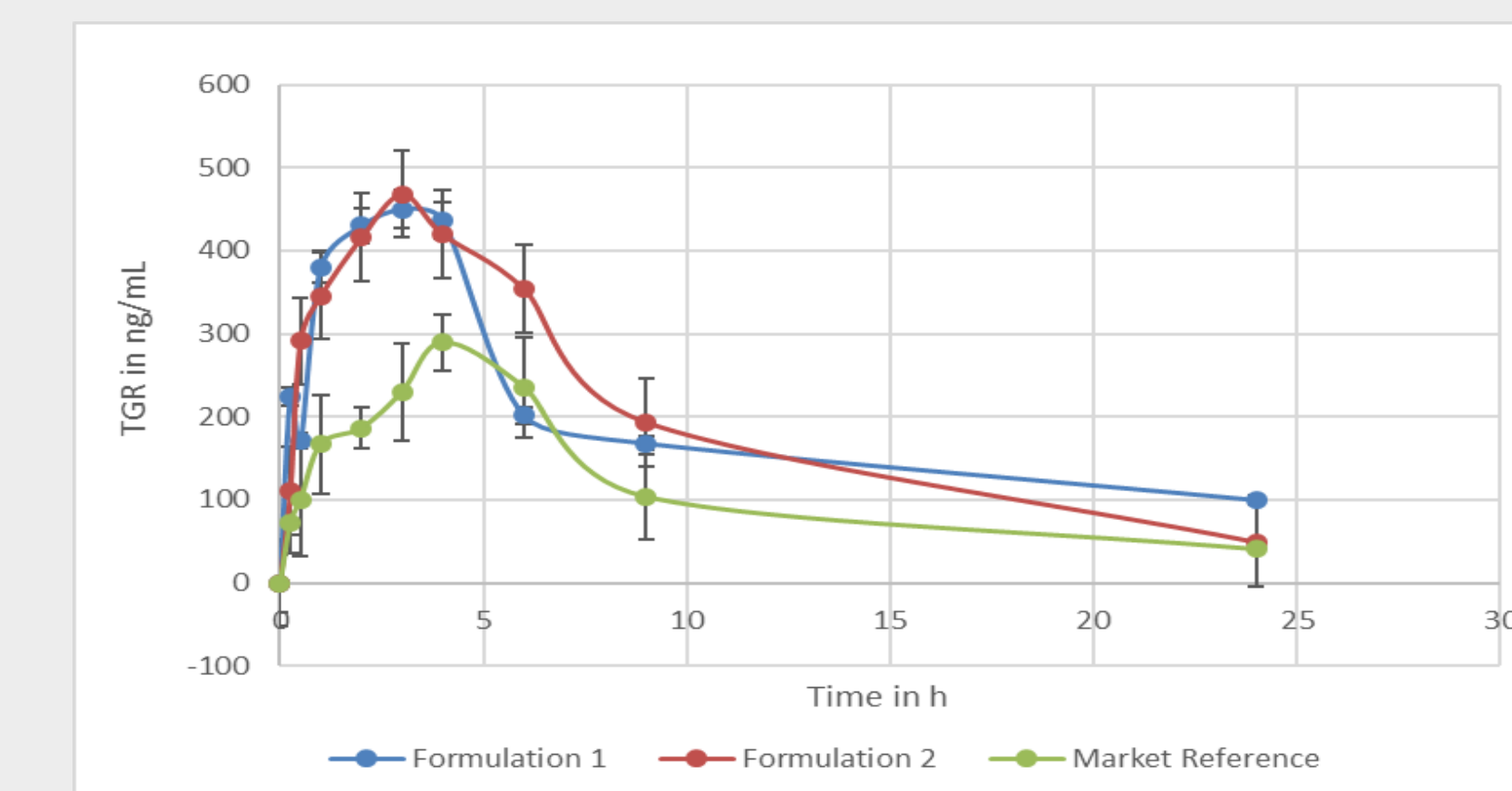


Fig. 4: *In vivo* Pharmacokinetic study- AUC.

In vivo performance of lipidic formulations was found to be almost 2 fold better compared to market reference (Table 2, Fig. 4).

CONCLUSIONS

- Gelucire 48/16® was employed effectively to enhance the solubility, dissolution and bioavailability enhancement of a poorly water-soluble BCS Class-IV drug i.e. TGR.
- In vitro* lipolysis study proved to be a good tool to predict the *in vivo* performance of lipidic formulations.
- The *in vivo* pharmacokinetic study on rat model showed significant improvement in the *in vivo* performance of formulations. The results obtained are in agreement with the results of *in vitro* lipolysis tests.
- It can be concluded that Gelucire 48/16® is an effective solubilizing excipient for solubility and dissolution enhancement of poorly soluble molecules.
- Gelucire 48/16® is can be applied for the solubility and bioavailability enhancement from super-generic (505 b ii) and product lifecycle management perspectives.

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